(a) Candidate Pool Budget K & Latency Trade-off

Candidate Recall@K (%)

Candidate Retrieval Latency (ms)

Candidate Pool Budget (K)

Pareto Knee ($K = 50$, 60.1%)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Latency (ms/sample)

(b) RRF Smoothing Parameter μ across City Densities

RRF Fused Top-10 Recall (%)

RRF Smoothing Constant (μ)

NYC Opt: $\mu = 30$ (62.9%)

CA Opt: $\mu = 30$ (44.2%)

CA (Sparse, Large-scale)
NYC (Compact, Dense)
TKY (High-density Network)

(c) Semantic Cluster Number K_c Trade-off

Long-tail Slice HR@10 (%)

Non-singleton Bucket Coverage (%)

Semantic Cluster Vocabulary Capacity (K_c)

Optimal $K_c = 256$ (Peak 42.5%)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Bucket Coverage (%)

(d) Profile Token Length L_{prof} Trade-off

End-to-End Retention HR@10 (%)

Valid JSON Parse Rate (%)

Max Profile Token Length (L_{prof})

Optimal $L_{prof} = 120$ (100% Parse, 42.5%)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Valid JSON Parse Rate (%)